C.3 Additional Concept Coverage Statistics

D LLM-based Autointerpretability

D.1 Prompts

Generate description and summary

Analyze this protein dataset to determine what predicts the 'Maximum activation value' and 'Amino acids of highest activated indices in protein' columns. This description should be as concise as possible but sufficient to predict these two columns on held-out data given only the description and the rest of the protein metadata provided. The feature could be specific to a protein family, a structural motif, a sequence motif, a functional role, etc. These WILL be used to predict how much unseen proteins are activated by the feature so only highlight relevant factors for this.
Focus on:

- Properties of proteins from the metadata that are associated with high vs medium vs low activation.
- Where in the protein sequence activation occurs (in relation to the protein sequence, length, structure, or other properties)
- What functional annotations (binding sites, domains, etc.) and amino acids are present at or near the activated positions
- This description that will be used to help predict missing activation values should start with "The activation patterns are characterized by:"

Then, in 1 sentence, summarize what biological feature or pattern this neural network activation is detecting. This concise summary should start with "The feature activates on"
Protein record: `Insert table with Swiss-Prot metadata and activation levels`

Predict activation levels

Given this protein metadata record, feature description, and empty table with query proteins, fill out the query table indicating the maximum feature activation value within in each protein (0.0-1.0).

Base activation value on how well the protein matches the described patterns. There could be 0, 1 or multiple separate instances of activation in a protein and each activation could span 1 or many amino acids.
Output only these values in the provided table starting with "Entry,Maximum activation value". Respond with nothing but this table.
Protein record: `Insert table with Swiss-Prot metadata`
Table to fill out with query proteins: `Insert empty table of IDs to fill out with predictions`
The activation patterns are characterized by: `Insert LLM description`

D.2 Example Descriptions and Summaries

Full Description Layer 4 Feature 9047

The activation patterns are characterized by:

- Highest activations (0.9-1.0) occur in glycosyltransferase proteins, particularly glycogen synthases (GlgA) and similar enzymes that transfer sugar molecules
- Activated positions consistently occur around amino acid positions 280-450 in these proteins, specifically involving glycine (G), alanine (A), or proline (P) residues
- The activated sites frequently overlap with substrate binding regions, particularly nucleotide-sugar binding sites (e.g., ADP-glucose, UDP-glucose, GDP-mannose)
- Medium activations (0.3-0.8) are seen in other transferases and synthetases with similar substrate binding patterns
- Proteins without sugar/nucleotide binding functions show no activation (0.0)

Feature	Pearson r	Feature Summary
4360	0.75	The feature activates on interchain disulfide bonds and surrounding hydrophobic residues in serine proteases, particularly those involved in venom and blood coagulation pathways.
9390	0.98	The feature activates on the conserved Nudix box motif of Nudix hydrolase enzymes, particularly detecting the metal ion binding residues that are essential for their nucleotide pyrophosphatase activity.
3147	0.70	The feature activates on conserved leucine and cysteine residues that occur in leucine-rich repeat domains and metal-binding structural motifs, particularly those involved in protein-protein interactions and signaling.
4616	0.76	The feature activates on conserved glycine residues in structured regions, with highest sensitivity to the characteristic glycine-containing repeats of collagens and GTP-binding motifs.
8704	0.75	The feature activates on conserved catalytic motifs in protein kinase active sites, particularly detecting the proton acceptor residues and surrounding amino acids involved in phosphotransfer reactions.
9047	0.80	The feature activates on conserved glycine/alanine/proline residues within the nucleotide-sugar binding domains of glycosyltransferases, particularly at positions known to interact with the sugar-nucleotide donor substrate.
10091	0.83	The feature activates on conserved hydrophobic residues (particularly V/I/L) within the catalytic regions of N-acetyltransferase domains, likely detecting a key structural or functional motif involved in substrate binding or catalysis.
1503	0.73	The feature activates on extracellular substrate binding loops of TonB-dependent outer membrane transporters, particularly those involved in nutrient uptake.
2469	0.85	The feature activates on conserved structural and sequence elements in bacterial outer membrane beta-barrel proteins, particularly around substrate binding and ion coordination sites in porins and TonB-dependent receptors.

Table 10: Example feature description summaries and their corresponding Pearson correlation coefficients when used along with more verbose descriptions to predict maximum activation levels.

E Additional cluster example

In 11 we see another example of proteins in common cluster united by a shared conceptual theme but varying in the specific proteins maximally activated and locations. Features 2244, 6262, and 9047 are not identified by a Swiss-Prot concept, but are maximally activated by specific sites within glycosyltransferases (labeled by examining examples at

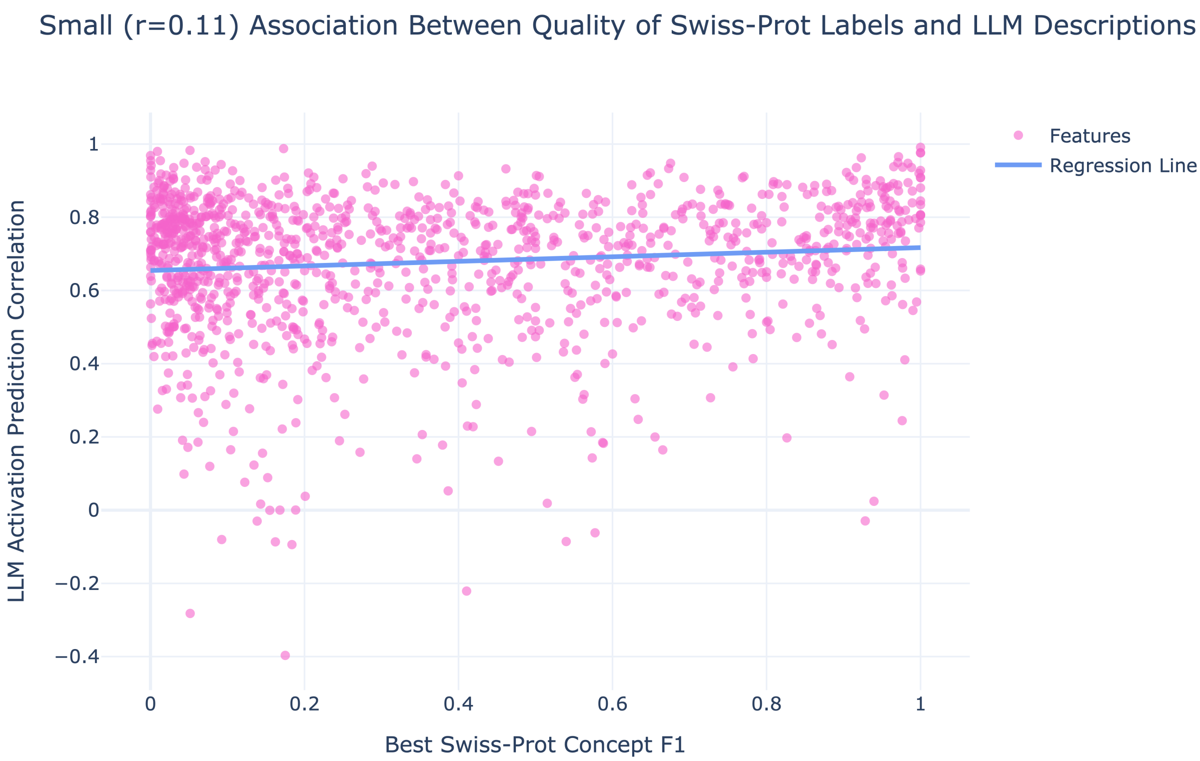

Figure 10: Comparing quality of Swiss-Prot concept labels and accuracy of LLM predicted feature activation patterns reveals small correlation.

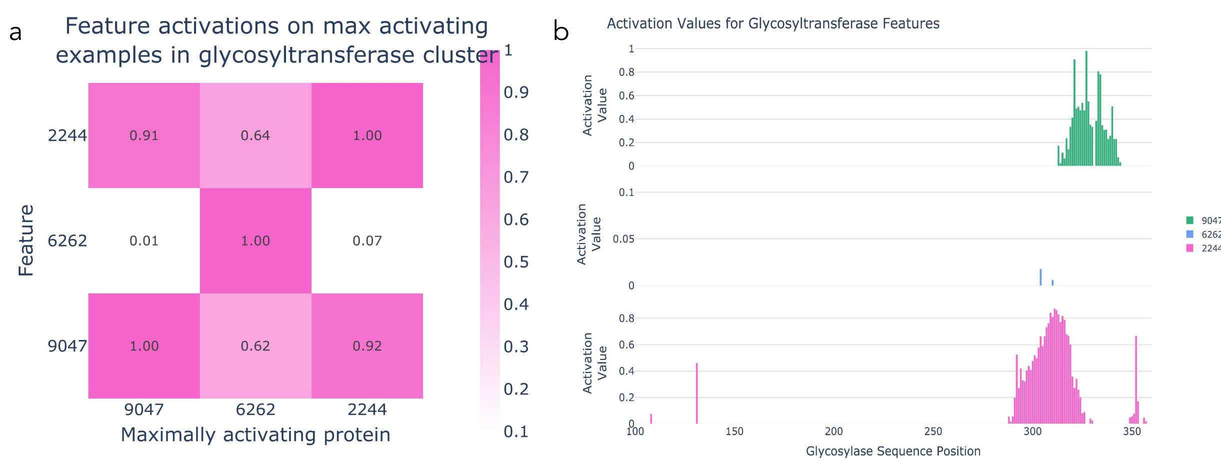

Figure 11: Example of features in glycosylastransferase cluster (2244,6262,9047) with varied maximally activating examples and maximally activating positions (a) Three glycosyltransferase features (y-axis) evaluated on the maximally activating proteins from each of these features (x-axis) (b) All features visualized on glycosyltransferase amsK (Uniprot: Q46638), maximally activating protein for 9047.